Blockchain Technology (BT) Complete Viva Questions & Answers

Unit I – Mathematical Foundation for Blockchain

Cryptography

Q1. What is Cryptography?

Ans. Cryptography is a technique used to protect information by converting it into a secure form so unauthorized people cannot read it.

Q2. What is Encryption?

Ans. Encryption converts readable data (plaintext) into unreadable data (ciphertext) using a key.

Q3. What is Decryption?

Ans. Decryption converts encrypted data back into its original readable form using a key.

Q4. What is Symmetric Key Cryptography?

Ans. Symmetric Key Cryptography uses the same secret key for both encryption and decryption.

Q5. What is Asymmetric Key Cryptography?

Ans. Asymmetric Key Cryptography uses two related keys: a public key and a private key.

Q6. Difference between Symmetric and Asymmetric Cryptography?

Ans. Symmetric uses one shared secret key and is generally faster. Asymmetric uses a public-private key pair and is generally slower.

Q7. Give examples of Symmetric Cryptography.

Ans. AES, DES, and 3DES are examples of symmetric encryption algorithms.

Q8. Give examples of Asymmetric Cryptography.

Ans. RSA and Elliptic Curve Cryptography (ECC) are examples of asymmetric cryptographic techniques.

Q9. What is ECC?

Ans. Elliptic Curve Cryptography (ECC) is a public-key cryptographic technique that provides strong security using relatively small keys.

Q10. Why is ECC used in Blockchain?

Ans. ECC is used for efficient digital signatures and public-key operations with smaller key sizes.

Hashing and Digital Signatures

Q11. What is a Cryptographic Hash Function?

Ans. A hash function converts input data of any size into a fixed-size hash value. It is designed to be one-way and collision-resistant.

Q12. What is SHA-256?

Ans. SHA-256 is a cryptographic hash algorithm that generates a 256-bit (32-byte) hash value.

Q13. Why is SHA-256 important in Blockchain?

Ans. SHA-256 helps verify data integrity and is used in Bitcoin's block hashing and Proof of Work.

Q14. What is Hashing?

Ans. Hashing converts input data into a fixed-length value called a hash using a hash function.

Q15. Is Hashing reversible?

Ans. No. Cryptographic hashing is designed to be one-way, so the original input cannot practically be recovered from its hash.

Q16. Difference between Hashing and Encryption?

Ans. Hashing is one-way and used for integrity. Encryption is reversible with the correct key and is used for confidentiality.

Q17. What is a Digital Signature?

Ans. A Digital Signature is created using a private key and verified using a public key to prove authenticity and integrity.

Q18. What is DSA?

Ans. Digital Signature Algorithm (DSA) is a public-key algorithm used to generate and verify digital signatures.

Q19. What is the purpose of a Digital Signature in Blockchain?

Ans. It verifies that a transaction was authorized by the private-key holder and was not modified.

Q20. What is a Merkle Tree?

Ans. A Merkle Tree is a tree structure where leaf nodes contain data hashes and parent nodes contain hashes of their children.

Q21. What is a Merkle Root?

Ans. The Merkle Root is the topmost hash of a Merkle Tree that represents all the included data.

Q22. Why are Merkle Trees used in Blockchain?

Ans. They allow efficient verification of transaction inclusion without downloading every transaction.

Unit II – Feature Engineering and Blockchain Fundamentals

Blockchain Basics

Q23. What is Blockchain?

Ans. Blockchain is a distributed digital ledger that stores records in cryptographically linked blocks.

Q24. Who introduced the Bitcoin Blockchain?

Ans. Satoshi Nakamoto published the Bitcoin whitepaper in 2008, and the network launched in 2009.

Q25. What is the main purpose of Blockchain?

Ans. To maintain a shared, verifiable, tamper-resistant record without relying on one central authority.

Q26. What is a Block?

Ans. A block is a collection of transactions or records along with information such as a timestamp and previous block hash.

Q27. What is a Genesis Block?

Ans. The Genesis Block is the first block of a blockchain.

Q28. What is a Distributed Ledger?

Ans. A Distributed Ledger is a shared database maintained across multiple computers or nodes.

Q29. What is Decentralization?

Ans. Decentralization means control and decision-making are distributed across multiple participants instead of one central authority.

Q30. What is a Node in Blockchain?

Ans. A node is a computer that participates in the blockchain network by maintaining, verifying, or sharing blockchain information.

Centralized vs Decentralized Systems

Q31. What is a Centralized System?

Ans. A Centralized System is controlled by a single organization or central server.

Q32. What is a Decentralized System?

Ans. A Decentralized System distributes control among multiple participating nodes.

Q33. Difference between Centralized and Decentralized Systems?

Ans. Centralized systems depend on one controlling authority. Decentralized systems distribute control and can reduce single points of failure.

Q34. What are the limitations of Centralized Systems?

Ans. Single point of failure, central control, censorship risk, and dependence on a trusted authority.

Q35. Why is Blockchain important?

Ans. Blockchain provides transparency, traceability, integrity, and shared trust among participants.

Q36. What is Immutability in Blockchain?

Ans. Immutability means confirmed blockchain records are extremely difficult to change without detection.

Layers of Blockchain

Q37. What are the five layers of Blockchain?

Ans. Application, Execution, Semantic, Propagation, and Consensus layers.

Q38. What is the Application Layer?

Ans. The Application Layer provides blockchain-based applications and services used by end users.

Q39. What is the Execution Layer?

Ans. The Execution Layer executes transactions and smart-contract instructions.

Q40. What is the Semantic Layer?

Ans. The Semantic Layer defines the rules, structure, and meaning of transactions and blockchain data.

Q41. What is the Propagation Layer?

Ans. The Propagation Layer handles communication and sharing of transactions and blocks between nodes.

Q42. What is the Consensus Layer?

Ans. The Consensus Layer helps network participants agree on the valid state and order of blockchain records.

Q43. What is Blockchain Adoption?

Ans. Blockchain Adoption means using blockchain in real-world applications such as finance, supply chains, healthcare, and identity management.

Unit III – Blockchain Platforms and Consensus

Blockchain Platforms

Q44. What are the main types of Blockchain?

Ans. Public, Private, and Consortium Blockchain.

Q45. What is Public Blockchain?

Ans. A Public Blockchain allows anyone to participate according to its network rules. Example: Bitcoin, Ethereum.

Q46. What is Private Blockchain?

Ans. A Private Blockchain restricts participation and access to authorized members.

Q47. What is Consortium Blockchain?

Ans. A Consortium Blockchain is governed by a group of organizations rather than one organization.

Q48. Difference between Public and Private Blockchain?

Ans. Public blockchains are generally open to anyone. Private blockchains restrict access to approved participants.

Q49. What is Bitcoin?

Ans. Bitcoin is a decentralized cryptocurrency network that allows peer-to-peer transfer of digital value.

Q50. What is Ethereum?

Ans. Ethereum is a blockchain platform that supports cryptocurrency transactions and programmable smart contracts.

Q51. What is Hyperledger?

Ans. Hyperledger is a collection of open-source enterprise blockchain projects hosted by the Linux Foundation.

Q52. What is Hyperledger Fabric?

Ans. Hyperledger Fabric is a permissioned blockchain framework used for enterprise business networks.

Q53. What is IOTA?

Ans. IOTA is a distributed ledger ecosystem designed for digital transactions and machine-to-machine applications, using a DAG-based architecture.

Q54. What is Corda?

Ans. Corda is a distributed ledger platform designed for business transactions between known parties.

Q55. What is R3?

Ans. R3 is the company associated with the development of the Corda distributed ledger platform.

Consensus Algorithms

Q56. What is Consensus in Blockchain?

Ans. Consensus is the process by which network participants agree on valid transactions and the shared ledger state.

Q57. Why is Consensus needed?

Ans. Consensus helps prevent conflicting records and ensures nodes follow a common set of rules.

Q58. What are Consensus Algorithms?

Ans. Consensus Algorithms are protocols that help distributed nodes agree on the ledger state.

Q59. What are important elements of Consensus?

Ans. Agreement, Validity, Termination, and Fault Tolerance are important properties of consensus.

Q60. What is Proof of Work (PoW)?

Ans. PoW requires miners to perform computational work to propose valid blocks. Example: Bitcoin.

Q61. What is Proof of Stake (PoS)?

Ans. PoS selects validators based on staked cryptocurrency and protocol rules rather than computational mining.

Q62. Difference between PoW and PoS?

Ans. PoW relies on computational work and consumes significant energy. PoS uses staked assets and generally consumes much less energy.

Q63. What is the Byzantine Generals Problem?

Ans. It describes the difficulty of achieving agreement among distributed participants when some may be faulty or dishonest.

Q64. What is Proof of Elapsed Time (PoET)?

Ans. PoET uses trusted execution environments to assign waiting times and select block producers.

Q65. What is Proof of Activity?

Ans. Proof of Activity is a hybrid consensus approach combining elements of Proof of Work and Proof of Stake.

Q66. What is Proof of Burn?

Ans. Proof of Burn requires participants to permanently destroy cryptocurrency to demonstrate commitment or obtain participation rights.

Q67. Which Consensus Algorithm does Bitcoin use?

Ans. Proof of Work (PoW).

Q68. Which Consensus Algorithm does Ethereum use?

Ans. Ethereum currently uses Proof of Stake (PoS).

Q69. What is a Validator?

Ans. A validator participates in checking transactions and proposing or confirming blocks in a PoS network.

Unit IV – Cryptocurrency, Bitcoin and Tokens

Cryptocurrency Basics

Q70. What is Cryptocurrency?

Ans. Cryptocurrency is a digital asset that uses cryptography and a distributed network to secure transactions.

Q71. What is Bitcoin?

Ans. Bitcoin (BTC) is a decentralized digital currency that allows peer-to-peer transactions without a central bank.

Q72. Who created Bitcoin?

Ans. Satoshi Nakamoto introduced Bitcoin in 2008.

Q73. What are the types of Cryptocurrency?

Ans. Common categories include Bitcoin, Altcoins, Stablecoins, and Privacy Coins.

Q74. What is an Altcoin?

Ans. An Altcoin is a cryptocurrency other than Bitcoin. Examples include Ether and Litecoin.

Q75. What is a Stablecoin?

Ans. A Stablecoin is a cryptocurrency designed to maintain a stable value relative to an asset such as the US dollar.

Q76. What is the difference between Coin and Token?

Ans. A coin is native to its blockchain, like ETH. A token is created using an existing blockchain, like an ERC-20 token on Ethereum.

Q77. What is Cryptocurrency Mining?

Ans. Mining uses computational power to create valid blocks in Proof-of-Work networks and earn rewards.

Q78. What is a Cryptocurrency Transaction?

Ans. A cryptocurrency transaction transfers digital assets or performs blockchain operations, usually authorized by a digital signature.

Q79. What are the uses of Cryptocurrency?

Ans. Digital payments, international transfers, decentralized finance, and blockchain application fees.

Cryptocurrency Wallets

Q80. What is a Crypto Wallet?

Ans. A crypto wallet manages cryptographic keys used to access and control cryptocurrency on a blockchain.

Q81. What is MetaMask?

Ans. MetaMask is a cryptocurrency wallet used to manage Ethereum-compatible accounts and interact with decentralized applications.

Q82. What is Coinbase?

Ans. Coinbase is a cryptocurrency exchange platform that provides services for buying, selling, and storing supported cryptocurrencies.

Q83. What is Binance?

Ans. Binance is a cryptocurrency exchange platform offering trading and other cryptocurrency-related services.

Q84. What is a Public Key?

Ans. A public key is a cryptographic key that can be shared and used to verify digital signatures.

Q85. What is a Private Key?

Ans. A private key is a secret cryptographic key used to authorize transactions by creating digital signatures.

Q86. What is a Wallet Address?

Ans. A wallet address is a public identifier used to receive cryptocurrency on a blockchain.

Q87. What is a Seed Phrase?

Ans. A seed phrase is a sequence of words used to recover a wallet's cryptographic keys. It must be kept secret.

Q88. What is a Hot Wallet?

Ans. A hot wallet is connected to an internet-enabled device. Example: MetaMask.

Q89. What is a Cold Wallet?

Ans. A cold wallet keeps private keys offline, reducing exposure to online attacks.

Q90. What is the difference between Hot and Cold Wallets?

Ans. Hot wallets are convenient for frequent transactions. Cold wallets provide stronger protection against online attacks.

Unit V – Ethereum Platform Using Solidity

Ethereum Fundamentals

Q91. What is Ethereum?

Ans. Ethereum is a decentralized blockchain platform used to execute smart contracts and build decentralized applications.

Q92. What is Ether?

Ans. Ether (ETH) is the native cryptocurrency of Ethereum, used for transaction fees and staking.

Q93. What is EVM?

Ans. Ethereum Virtual Machine (EVM) is the execution environment that runs smart-contract bytecode on Ethereum-compatible networks.

Q94. What is the purpose of EVM?

Ans. The EVM executes smart contracts consistently across Ethereum network nodes.

Q95. What are Ethereum Networks?

Ans. Ethereum has a Mainnet for real transactions and test networks for development and testing.

Q96. What is Ethereum Mainnet?

Ans. Mainnet is the live Ethereum network where transactions use real ETH.

Q97. What is an Ethereum Testnet?

Ans. A testnet allows developers to test smart contracts using test ETH without spending real ETH.

Q98. What is Sepolia?

Ans. Sepolia is an Ethereum test network used for testing smart contracts and decentralized applications.

Smart Contracts and Solidity

Q99. What is a Smart Contract?

Ans. A smart contract is a program stored on a blockchain that executes according to predefined rules.

Q100. What is Solidity?

Ans. Solidity is a high-level programming language used to develop smart contracts for Ethereum-compatible blockchains.

Q101. Why is Solidity used?

Ans. Solidity is used to write blockchain programs for transactions, tokens, voting, and other decentralized applications.

Q102. What are the types of Smart Contracts?

Ans. Common categories include legal smart contracts, DAO contracts, and application-logic contracts.

Q103. What is Smart Contract Deployment?

Ans. Deployment is the process of publishing compiled smart-contract code to a blockchain.

Q104. What is Remix IDE?

Ans. Remix IDE is a browser-based development environment used to write, compile, test, and deploy Solidity smart contracts.

Q105. What is Truffle?

Ans. Truffle is a development framework used to compile, test, and deploy Ethereum smart contracts.

Q106. What is Gas in Ethereum?

Ans. Gas is a unit used to measure the computational resources required to execute operations on Ethereum.

Q107. What is Gas Limit?

Ans. Gas Limit is the maximum amount of gas allowed for a transaction.

Q108. What is Gas Price?

Ans. Gas Price is the amount of ETH paid per unit of gas. Ethereum uses base fees and optional priority fees.

Q109. What is Gwei?

Ans. Gwei is a small unit of Ether. 1 ETH = 10^9 Gwei.

Q110. What is a Transaction Hash?

Ans. A transaction hash is a unique cryptographic identifier used to track a blockchain transaction.

Q111. What is a DApp?

Ans. A Decentralized Application (DApp) is an application that interacts with smart contracts or decentralized networks.

Swarm and Whisper

Q112. What is Swarm?

Ans. Swarm is a decentralized storage and content-distribution system associated with the Ethereum ecosystem.

Q113. What is Whisper?

Ans. Whisper is a historical Ethereum peer-to-peer messaging protocol designed for decentralized communication.

Q114. Difference between Swarm and Whisper?

Ans. Swarm is designed for decentralized data storage, while Whisper was designed for decentralized messaging.

Unit VI – Blockchain Case Studies

Q115. What are Blockchain Applications?

Ans. Blockchain is used in banking, healthcare, supply chains, government, IoT, and energy systems.

Q116. How is Blockchain used in Banking?

Ans. Blockchain can support cross-border payments, settlement, and shared transaction records.

Q117. How is Blockchain used in Retail?

Ans. Blockchain can improve product traceability, supply-chain transparency, and authenticity verification.

Q118. How is Blockchain used in Healthcare?

Ans. Blockchain can maintain tamper-evident audit records and controlled sharing of medical-data references.

Q119. How is Blockchain used in Government?

Ans. Blockchain can support record verification, digital identity, land records, and transparent audit trails.

Q120. How is Blockchain used in IoT?

Ans. Blockchain can provide device identity, secure event records, and decentralized coordination between IoT devices.

Q121. How is Blockchain used in Energy?

Ans. Blockchain can support peer-to-peer energy trading, renewable-energy tracking, and automated settlement.

Q122. What is Blockchain Integration?

Ans. Blockchain Integration means connecting blockchain with existing applications, databases, or technologies.

Q123. What is Blockchain with AI?

Ans. Blockchain and AI can combine tamper-evident data records with intelligent analysis and automated decisions.

Q124. What is Blockchain with Cloud Computing?

Ans. Cloud Computing provides infrastructure for hosting blockchain nodes, applications, and supporting services.

Q125. What are the limitations of Blockchain?

Ans. Major limitations include scalability, transaction costs, privacy challenges, storage requirements, and regulatory uncertainty.

Group C – Blockchain Technology Practical Viva

All 6 practical assignments from your syllabus

Practical 1 – MetaMask Installation and Ether Transaction Fees

Q126. What is MetaMask?

Ans. MetaMask is a cryptocurrency wallet used to manage Ethereum-compatible accounts and connect with blockchain applications.

Q127. Why do we install MetaMask?

Ans. To create a wallet, manage accounts, sign transactions, and interact with smart contracts.

Q128. What are the steps to install MetaMask?

Ans.

1. Install MetaMask from its official source.
2. Create a new wallet.
3. Securely save the recovery phrase.
4. Set a password and open the wallet.

Q129. What is Gas Fee?

Ans. A Gas Fee is the transaction fee paid for computational work on the Ethereum network.

Q130. Why do we pay Gas Fees?

Ans. Gas fees compensate network validators and help prevent misuse of network resources.

Q131. What happens if a transaction runs out of Gas?

Ans. The transaction fails or reverts, but the gas consumed is still charged.

Q132. What is the formula for Ethereum Transaction Fee?

Ans. Transaction Fee = Gas Used × Effective Gas Price.

Q133. What is Etherscan?

Ans. Etherscan is an Ethereum blockchain explorer used to view transactions, wallet addresses, gas fees, and smart contracts.

Practical 2 – Creating a Crypto Wallet

Q134. How do you create a Wallet using MetaMask?

Ans.

1. Open MetaMask.
2. Select Create Wallet.
3. Secure the recovery phrase.
4. Set up the wallet password.

Q135. What is the purpose of a Wallet Address?

Ans. A wallet address is used to receive cryptocurrency and identify an account on the blockchain.

Q136. What is the difference between Public Key and Private Key?

Ans. A public key can be shared for verification, while a private key must remain secret and is used to sign transactions.

Q137. What happens if you lose your Seed Phrase?

Ans. If you lose your recovery phrase and all other access methods, you may permanently lose access to your cryptocurrency.

Q138. Can we create multiple accounts in MetaMask?

Ans. Yes. MetaMask supports multiple accounts within a wallet.

Q139. Can we use MetaMask on a Test Network?

Ans. Yes. MetaMask supports compatible test networks such as Sepolia.

Practical 3 – Bank Account Smart Contract

Q140. What is the aim of this practical?

Ans. To create a Solidity smart contract that performs deposit, withdrawal, and balance-checking operations.

Q141. Which programming language is used?

Ans. Solidity.

Q142. What is a Deposit Function?

Ans. A deposit function receives Ether and updates the balance stored or tracked by the smart contract.

Q143. What is a Withdraw Function?

Ans. A withdraw function transfers Ether from the smart contract according to the contract's balance and authorization rules.

Q144. What is a Balance Function?

Ans. A balance function returns the available balance stored or tracked by the smart contract.

Q145. What is the payable keyword?

Ans. payable allows a Solidity function or address to receive Ether.

Q146. What is msg.sender?

Ans. msg.sender represents the immediate caller of a smart-contract function.

Q147. What is msg.value?

Ans. msg.value represents the amount of Ether, in wei, sent with a function call.

Q148. What is require() in Solidity?

Ans. require() checks a condition and reverts the transaction if the condition is false.

Q149. What is the view keyword?

Ans. view indicates that a function reads blockchain state without modifying it.

Q150. What is the difference between view and pure?

Ans. view functions can read contract state. pure functions cannot read or modify contract state.

Practical 4 – Student Data Using Solidity

Q151. What is the aim of the Student Data Practical?

Ans. To create a Solidity smart contract that stores student information using structures and arrays and demonstrates a fallback function.

Q152. What is a Structure in Solidity?

Ans. A struct is a user-defined data type that groups related variables under one name.

Q153. Why do we use Structures?

Ans. To store related data such as student ID, name, and marks together.

Q154. What is an Array in Solidity?

Ans. An array stores multiple elements of the same data type.

Q155. What is a Dynamic Array?

Ans. A dynamic array can grow in size as elements are added.

Q156. What is a Fallback Function?

Ans. A fallback function executes when a contract receives a call that does not match an existing function, or under certain other call conditions.

Q157. What is the receive() Function?

Ans. receive() is a special payable function that handles plain Ether transfers with empty calldata.

Q158. What is the difference between fallback() and receive()?

Ans. receive() handles plain Ether transfers. fallback() handles unmatched function calls and can receive Ether if payable.

Q159. What is the storage keyword?

Ans. storage refers to persistent blockchain data stored in a smart contract.

Q160. What is the memory keyword?

Ans. memory refers to temporary data used during function execution.

Q161. What is the difference between Storage and Memory?

Ans. Storage keeps persistent contract data, while Memory holds temporary data during execution.

Q162. Why do we observe Gas Values?

Ans. To understand the computational cost and transaction fees of deploying and executing smart contracts.

Practical 5 – Survey of Blockchain Types and Use Cases

Q163. What is the aim of this practical?

Ans. To study different types of blockchains and compare their real-world applications.

Q164. What are the three main types of Blockchain?

Ans. Public, Private, and Consortium Blockchain.

Q165. Give an example of Public Blockchain.

Ans. Bitcoin and Ethereum.

Q166. Give an example of Private Blockchain.

Ans. A Hyperledger Fabric network controlled by a single organization.

Q167. Give an example of Consortium Blockchain.

Ans. A Hyperledger Fabric network jointly governed by multiple banks or organizations.

Q168. Which Blockchain is suitable for Banking?

Ans. A permissioned blockchain is often suitable because banks need controlled access, privacy, and compliance.

Q169. Which Blockchain is suitable for Cryptocurrency?

Ans. Public blockchain, because it supports open participation and decentralized transactions.

Practical 6 – Business Network Using Hyperledger

Q170. What is the aim of this practical?

Ans. To create a business network using Hyperledger for secure transactions and data sharing between authorized participants.

Q171. What is Hyperledger Fabric?

Ans. Hyperledger Fabric is an open-source permissioned blockchain framework for enterprise applications.

Q172. What is a Permissioned Blockchain?

Ans. A permissioned blockchain allows only authorized participants to access or perform specified network operations.

Q173. What is Chaincode?

Ans. Chaincode is the smart-contract implementation used in Hyperledger Fabric to define business logic.

Q174. What is a Peer in Hyperledger?

Ans. A peer is a network node that maintains ledger data and may execute or endorse transactions.

Q175. What is an Ordering Service?

Ans. The Ordering Service establishes transaction order and delivers blocks to peers.

Q176. What is a Channel in Hyperledger Fabric?

Ans. A channel provides a separate communication and ledger environment for selected network organizations.

Q177. What is MSP?

Ans. Membership Service Provider (MSP) defines trusted identities and membership rules in a Hyperledger Fabric network.

Q178. What is the difference between Ethereum and Hyperledger Fabric?

Ans. Ethereum supports public, permissionless blockchain applications. Hyperledger Fabric is designed for permissioned enterprise networks.

Q179. Does Hyperledger Fabric require cryptocurrency?

Ans. No. Hyperledger Fabric does not require a native cryptocurrency for transaction processing.

Q180. What are the benefits of Hyperledger?

Ans. Controlled access, privacy, modular architecture, and enterprise transaction management.

Last-Minute Blockchain Practical Revision

Practical	Important terms to remember
1. MetaMask Installation	Gas, Ether, Gwei, Transaction Fee
2. Wallet Creation	Public Key, Private Key, Seed Phrase
3. Bank Smart Contract	payable, msg.sender, msg.value, require()
4. Student Data	struct, array, fallback, storage, memory
5. Blockchain Survey	Public, Private, Consortium
6. Hyperledger	Chaincode, Peer, Channel, MSP